R13

Code No: 115AP

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, August - 2022
COMPILER DESIGN
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) What is LEX? Explain in brief about the different sections of LEX Program.
b) Differentiate between a compiler and an interpreter. [10+5]

2.a) Define a Compiler. Explain in brief about the types of lexical errors with example.
b) Explain in brief about compiler construction tools. [7+8]

3. Construct CLR Parsing table for the grammar: [15]
S→L = R ¦ R
L→ * R ¦ id
R→L

4. Demonstrate various conflicts raised in shift-reduce parsers with an example. [15]

5.a) Construct an annotated parse tree for 3*4+5.
b) Translate the expression –(a+b)*(c+d)+(a+b+c) in to quadruple, triple and indirect triple.
c) What are attributed grammars? Give an example. [4+6+5]

6. Define Symbol table. Explain about the data structures used for Symbol table. [15]

7. Define code generator. Discuss about Instruction Selection and Register allocation. [15]

8.a) Explain in brief about different Principal sources of optimization techniques with suitable examples.
b) What is a Flow Graph? Explain how a given program can be converted in to a Flow graph. [7+8]

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